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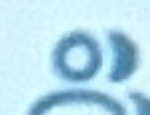
```
# Lab 14: Implement a Pre-trained CNN model as a Feature Extractor
# =====

import torch
import torch.nn as nn
import torch.optim as optim
from torchvision import models, datasets, transforms
from torch.utils.data import DataLoader
import matplotlib.pyplot as plt
import os

# -----
# Step 1: Define Transformations
# -----
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
])
```


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```
# Download a small example dataset (CIFAR-10)
train_data = datasets.CIFAR10(root='./data', train=True, download=True, transform=transform)
test_data = datasets.CIFAR10(root='./data', train=False, download=True, transform=transform)

train_loader = DataLoader(train_data, batch_size=32, shuffle=True)
test_loader = DataLoader(test_data, batch_size=32, shuffle=False)

# -----
# Step 2: Load Pre-trained Model
# -----
model = models.resnet18(pretrained=True)

# Freeze all convolutional layers
for param in model.parameters():
    param.requires_grad = False

# -----
# Step 3: Replace the Classifier
```



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```
num_features = model.fc.in_features
model.fc = nn.Linear(num_features, 10) # CIFAR-10 has 10 classes

# Move model to device
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)

# -----
# Step 4: Define Loss and Optimizer
# -----
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.fc.parameters(), lr=0.001)

# -----
# Step 5: Train Only the Classifier
# -----
num_epochs = 2 # keep small for quick run
train_loss_list = []
```

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```
[1]  for epoch in range(num_epochs):
    running_loss = 0.0
    for images, labels in train_loader:
        images, labels = images.to(device), labels.to(device)

        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item()

    avg_loss = running_loss / len(train_loader)
    train_loss_list.append(avg_loss)
    print(f"Epoch [{epoch+1}/{num_epochs}], Loss: {avg_loss:.4f}")

print("\n Training completed successfully!")

# -----
```


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[1]▶plt.plot(train_loss_list, marker='o')plt.title("Training Loss per Epoch")plt.xlabel("Epoch")plt.ylabel("Loss")plt.grid(True)plt.show()

Step 7: Test Accuracy

correct = 0
total = 0
model.eval()
with torch.no_grad():
 for images, labels in test_loader:
 images, labels = images.to(device), labels.to(device)
 outputs = model(images)
 _, predicted = torch.max(outputs, 1)
 total += labels.size(0)

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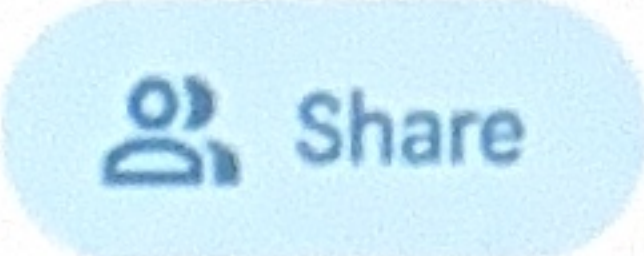
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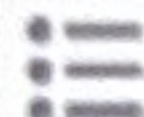
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```
_, predicted = torch.max(outputs, 1)
total += labels.size(0)
correct += (predicted == labels).sum().item()

accuracy = 100 * correct / total
print(f"\n Test Accuracy: {accuracy:.2f}%")
```

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/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:208: UserWarning: The parameter 'pretrained' is deprecated
warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:223: UserWarning: Arguments other than a weight enum or
warnings.warn(msg)
Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to /root/.cache/torch/hub/checkpoints/resnet18-f3707
100%|██████████| 44.7M/44.7M [00:00<00:00, 156MB/s]
Epoch [1/2], Loss: 0.8107
Epoch [2/2], Loss: 0.6423

✓ Training completed successfully!

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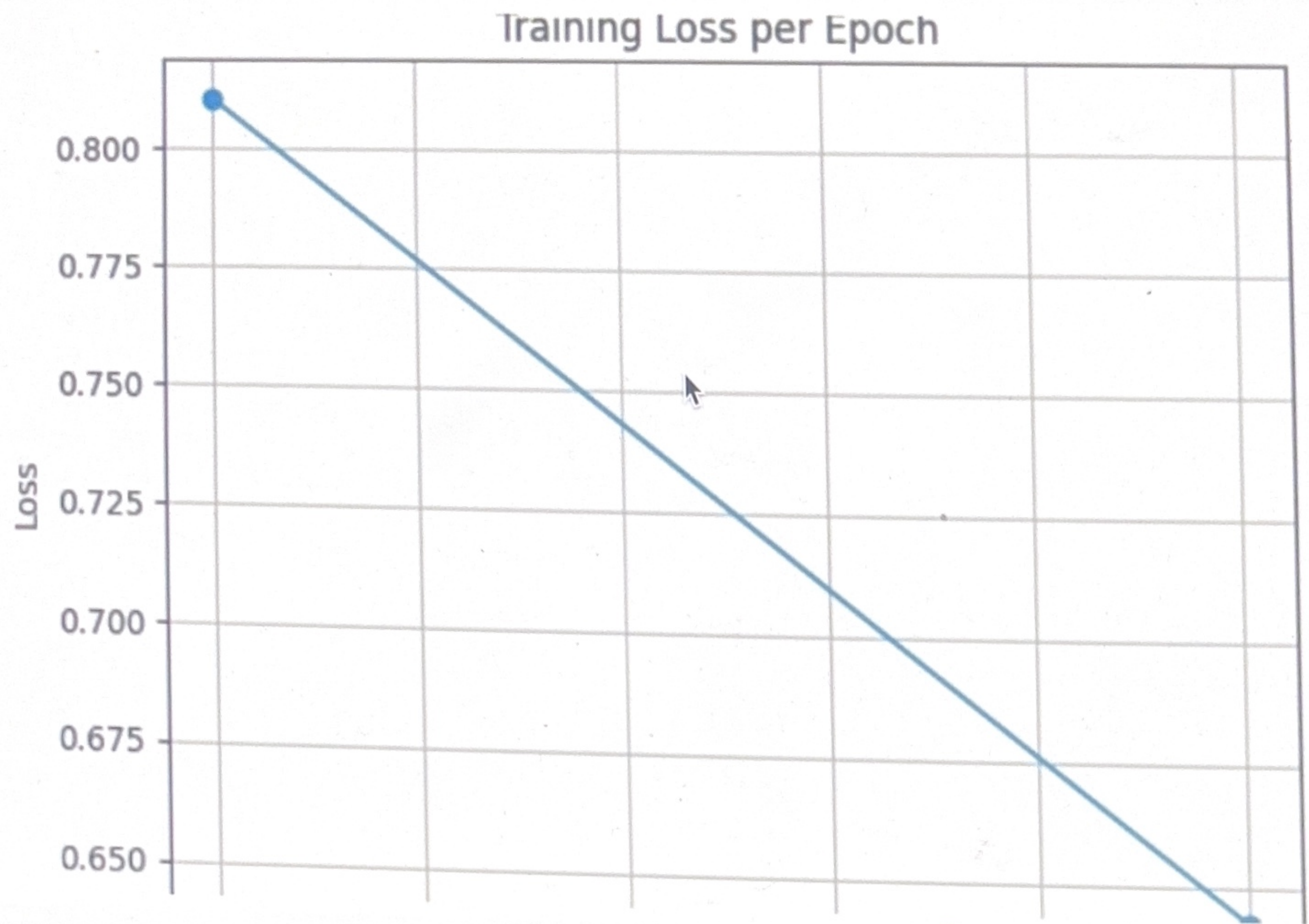


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Epoch	Test Accuracy
0.58	0.70
1.0	0.65

Test Accuracy: 79.21%

[]

Start coding or generate with AI.

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ENG 16:05

14. Implement a pre-trained CNN model as feature extractor.

Aim:-

To implement a pre-trained CNN as feature extractor using transfer learning.

Objective:-

- To understand how a CNN pre-trained on ImageNet can extract visual feature
- To freeze convolution layer and use their output feature for new tasks.
- To train only the classifier head for faster and more accurate learning.
- To observe how transfer learning improves performance with limited data.

pseudo code:-

1. Import torchvision, model and torch libraries.
2. Load a pretrained CNN from torchvision.
3. Freeze all convolutional layer to prevent retraining.
4. Replace the final classifier layer for new number of classes.
5. Load a small dataset (eg. CIFAR-10).
6. Extract features using pretrained backbone.
7. Train only the new classifier head.
8. Evaluate accuracy and visualize results.

Observation:-

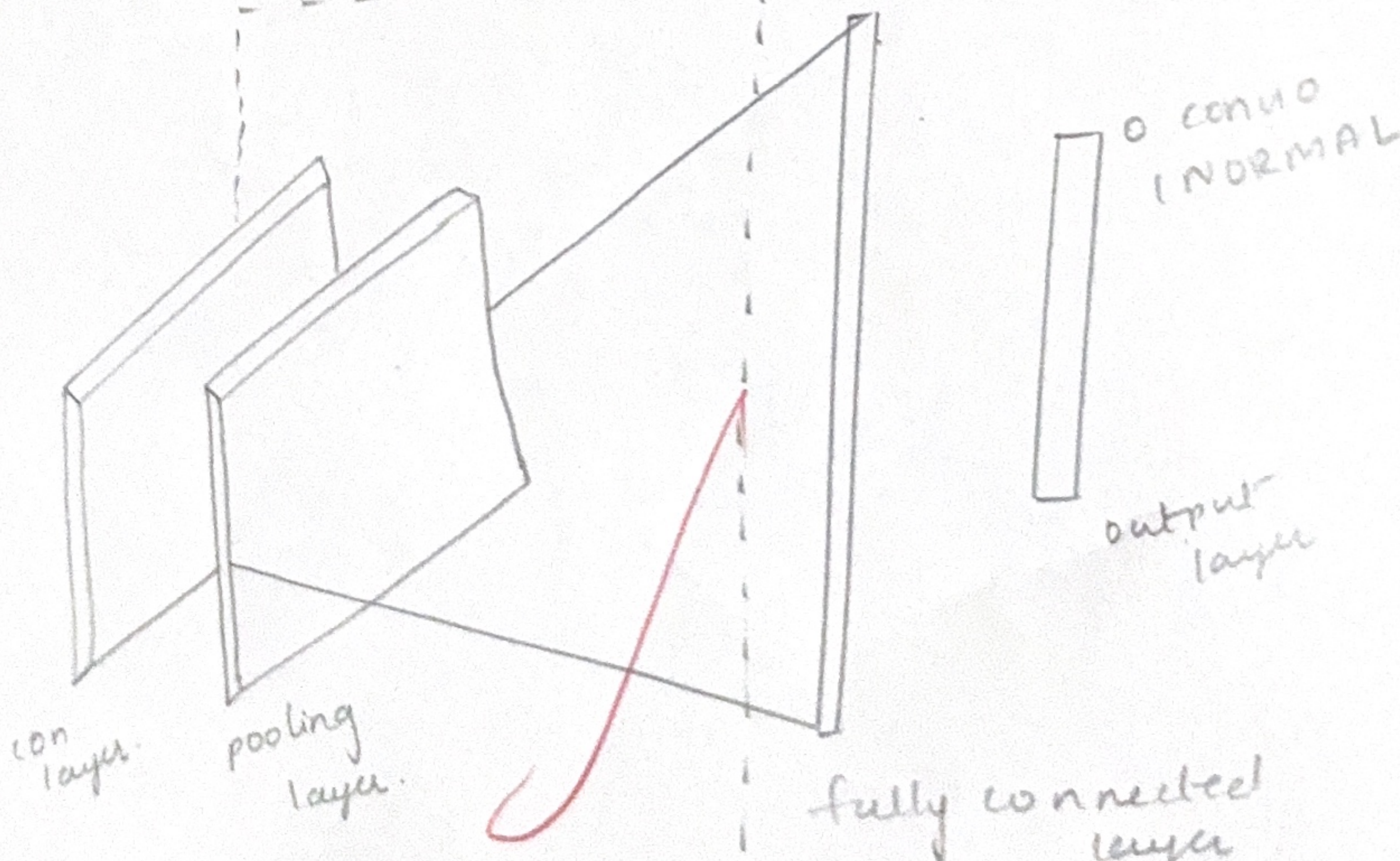
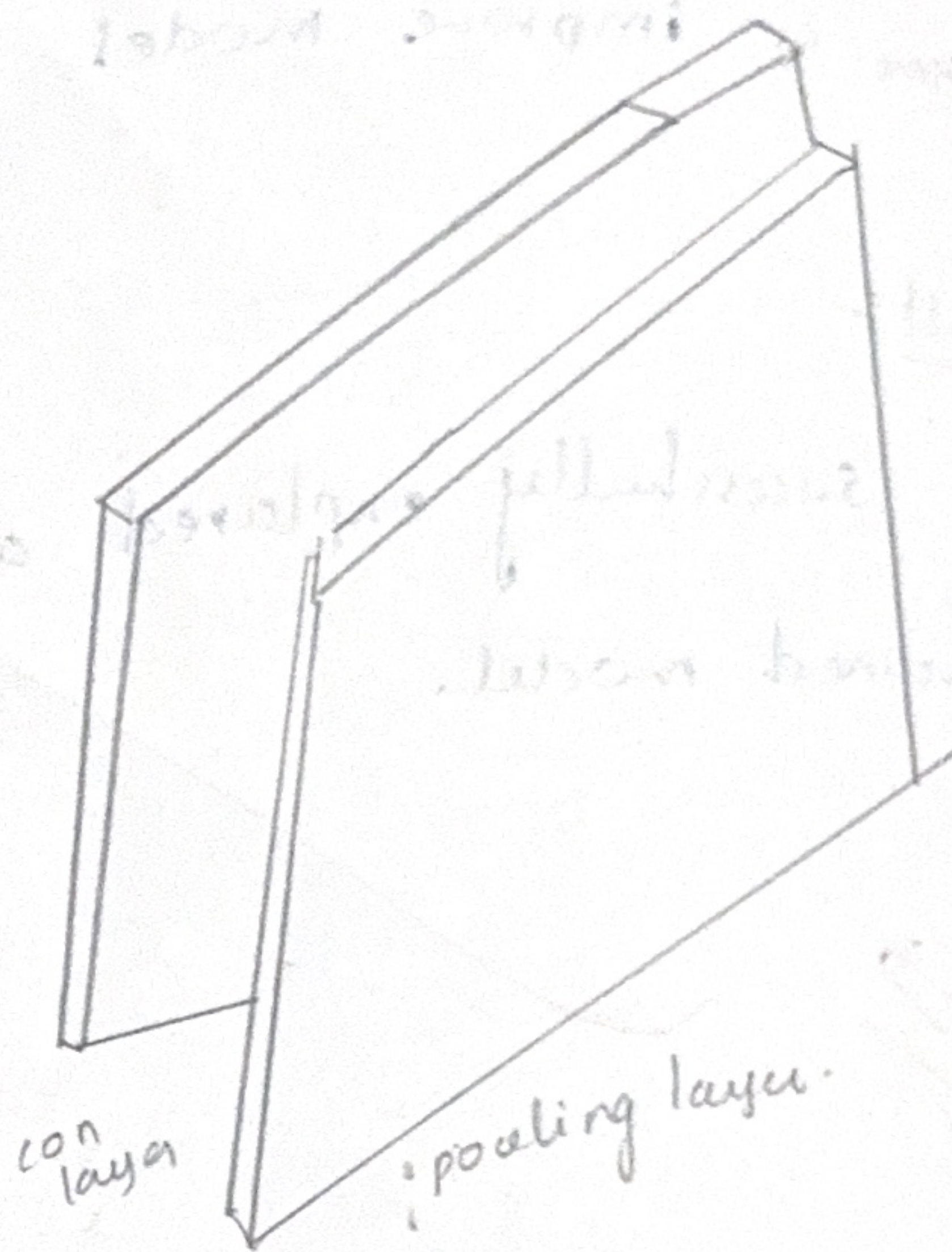
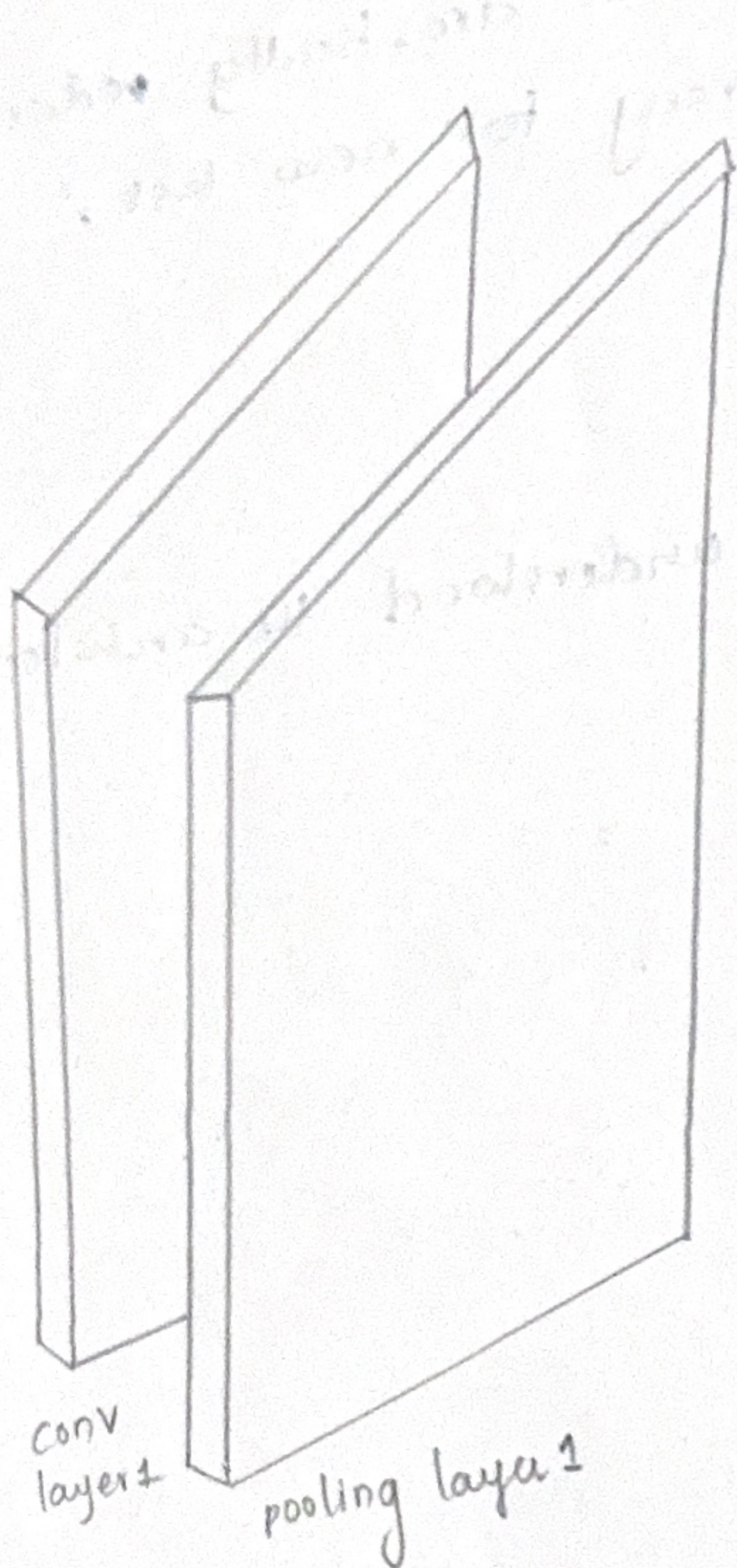
Aspect	ResNet feature Extract	Training from scratch.
Training time	very low	high
Accuracy	- 85%	- 65%
Required & data	Small	large
Key benefit	Reuses pre-trained visual feature	must learn everything new.

Result:-

Successfully implemented a pre-trained CNN as a feature Extractor.

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pre-trained CNN model



feature extractor

classifier

freeze pretrained CNN architecture

retrain